

# Non-One-to-One Cross-Chain Forensic Fund-Flow Correspondence: Transport Representation and Conditional Decoding

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**Abstract**—Cross-chain money laundering fragments suspicious asset trails across bridges, tokens, and blockchains. Existing transaction-level tracing tools struggle with splitting, merging, bridge latency, amount drift, and incomplete event evidence, which makes one-to-one transaction matching fragile for operational anti-money-laundering (AML) analysis. The core observation of this paper is that fund-flow correspondence in cross-chain forensics is not always one-to-one: one source-chain flow may correspond to several destination-chain flows, and several source flows may consolidate into one destination flow. Existing transaction-level matchers answer “which source transaction corresponds to which destination transaction,” and their one-to-one representation mismatches this form of fund reorganization. We therefore formalize Cross-Chain Suspicious Fund-Flow Correspondence (CSFFC) as a flow-level, mass-bearing soft-correspondence problem, and propose RC-UOT, a risk-constrained unbalanced optimal transport model, whose nonnegative transport plan uniformly expresses one-to-one, one-to-many, many-to-one, and unmatched mass, and encodes amount consistency, temporal causality, path consistency, risk consistency, and evidence quality into a composite cost with forensic semantics. Further analysis shows that raw transport-plan values are not pure correspondence affinities: their rankings are distorted by transport scaling and marginal pressure. We propose directional conditional decoding—row-direction scores normalized by column mass and column-direction scores normalized by row mass, decoded with a mutual top- $k$  rule—together with its dual-cancellation property: the row-direction conditional score removes the destination-side dual scaling exactly, and the column-direction score removes the source-side dual scaling exactly. This identity is stated as a direct corollary of standard Sinkhorn structure; we do not claim it as a new normalization. In a preregistered, hash-locked, one-shot confirmatory holdout on untouched seeds across three bridges (Celer, Multichain, PolyNetwork), with an independent verifier recomputing every result, conditional decoding raises macro edge F1 from 0.2350 for raw-plan decoding to 0.3104, with primary effect  $\Delta = +0.075413$  and paired 95% CI [0.070557, 0.080312], positive on every bridge, and all five preregistered mechanism-direction indicators in the repair direction; the same experiment yields a balanced-versus-unbalanced conditional-decoding difference of  $-0.0003$  with a CI including zero, so no unbalanced-relaxation-specific ranking gain is claimed. For structural representation, controlled semi-synthetic stress tests show that the transport representation can express  $1 \rightarrow N$  edge sets that one-to-one decoders cannot represent (edge-inclusion recall, not exact topology recovery); we also report transparently that strict exact-topology recovery is zero for every tested method at the frozen decode, and that a globally calibrated threshold rule dominates edge F1. An independent post-development temporal data audit (data-only; no method executed) confirms that real protocol-grounded flow-level fan-out structure exists outside the development corpus (2,451 Tier-A source-level fan-out units over a 359.98-day span), answering problem validity; because the corpus did not meet the preregistered adequacy gate ( $G = 7 < 8$ ), no method

was executed on it and no external-performance claim is made. Method performance on an adequately diverse independent temporal corpus remains unestablished.

**Index Terms**—Cross-chain forensics, anti-money laundering, fund-flow correspondence, non-one-to-one correspondence, optimal transport, conditional decoding, selective abstention.

## I. INTRODUCTION

### A. Problem and Motivation

Cross-chain bridges, decentralized exchanges, and multi-hop routing let funds under investigation be split, swapped, released with delay, and re-aggregated at new addresses across chains, and regulators now flag chain-hopping and cross-bridge typologies as elevated virtual-asset laundering risks [1]–[3]. Cross-chain AML investigation is therefore a *fund-flow correspondence* problem spanning ledgers, protocols, asset forms, and time windows. Fan-out or merge structure is not, in itself, evidence of illicit intent—it can arise from legitimate routing, aggregation, or liquidity management—so this paper addresses evidence-constrained forensic correspondence, not automatic illicitness classification.

Transaction-level cross-chain tracers such as CONNECTOR [4] and ABCTracer [5] are effective when bridge events are explicit, but they exhibit structural mismatches with cross-chain fund reorganization: one source activity may *fan out* to several destinations or several activities may *merge* into one ( $1 \rightarrow N / N \rightarrow 1$ ), bridge latency breaks naive hash-level alignment, fees cause amount drift, and event evidence is often incomplete. Hard one-to-one matching cannot represent soft mass allocation, and a high hit rate on predominantly one-to-one anchors does not imply robustness under fan-out/merge stress. CSFFC is designed for these conditions rather than to replace transaction-level tools (Fig. 1).

### B. CSFFC Formulation

**CSFFC (Cross-Chain Suspicious Fund-Flow Correspondence)** formalizes soft, mass-bearing correspondence between source-chain and destination-chain fund flows. A *fund flow* is the AML observation unit: bridge-relevant activity aggregated by primary address, chain, asset context, and a rolling 1800-s window; flows are not native on-chain atoms but preserve amount, timing, route, and evidence references [6]. CSFFC seeks a nonnegative transport plan  $P$  in which  $P_{ij}$  allocates mass from source flow  $s_i$  to destination flow  $t_j$ . Labels are directed flow edges with pattern metadata; the frozen label layer’s legacy field names are inverted relative to canonical topology (legacy `many_to_one` =  $1 \rightarrow N$  fan-out, legacy `one_to_many` =  $N \rightarrow 1$  merge; Table II footnote), and this paper uniformly uses fan-out =  $1 \rightarrow N$ , merge =  $N \rightarrow 1$ .

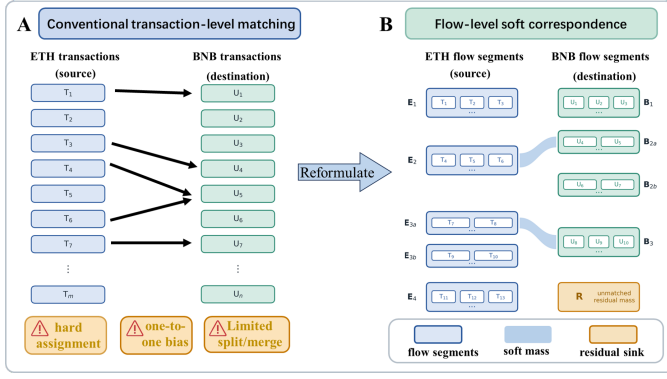


Fig. 1. Reformulation from transaction-level one-to-one matching to flow-level soft correspondence: transaction hashes, event logs, address interactions, amounts, and temporal relations are organized into auditable fund-flow evidence chains, and soft correspondence uniformly expresses one-to-one matches, fan-out, merge, and unmatched residuals.

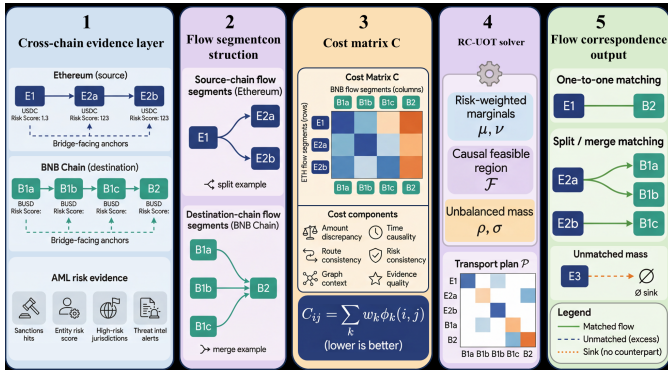


Fig. 2. RC-UOT framework: source/destination transaction logs, bridge events, and AML risk evidence are organized into fund-flow segments; the composite cost matrix is solved by RC-UOT into a soft transport plan expressing one-to-one, fan-out/merge, and unmatched mass.

### C. Method and Tool Overview

**RC-UOT (Risk-Constrained Unbalanced Optimal Transport)** solves CSFFC with entropy-regularized unbalanced optimal transport [7]–[9]: pairwise costs combine amount error, temporal causality, bridge/path consistency, AML risk, and evidence quality; unbalanced marginals allow *unmatched mass* (partial observation) instead of forced full assignment. The soft plan is decoded into reported edges by a *directional conditional decoder* (Section III-C) that repairs the ranking distortion introduced by the raw plan’s dual scalings. **RC-UOT-Q** adds *event-backed quotient-space grouping* and *coverage qualification* (tiers A/B/C/Uncovered) with *selective abstention* when evidence is insufficient—an AML safety mechanism (Fig. 2). We release **Cross AML** (tools/cross\_aml/), a standalone open-source prototype for evidence-qualified cross-chain AML triage that collects RPC evidence, constructs flows and quotient groups, runs RC-UOT-Q-style matching with abstention, and emits interpretable reports.

### D. Contributions

(1) **Non-one-to-one forensic formulation.** CSFFC: flow-level, mass-bearing correspondence with explicit unmatched mass, partial observability, and coverage-qualified abstention, distinguishing protocol-level one-to-one settlement from flow-level fan-out/merge structure. (2) **Transport representation and conditional decoding.** We diagnose that raw transport-plan values mix correspondence affinity with transport scaling and marginal pressure; directional conditional decoding with an exact dual-cancellation analysis (Proposition 1) repairs this ranking distortion, presented as an adaptation of standard mutual-matching normalization, not a new normalization. (3) **Faithful structural-capability evaluation.** A code-level capability audit and a unified strict evaluation across Celer, Multichain, and PolyNetwork establish what one-to-one decoders structurally cannot represent and bound what the transport decode achieves (edge-inclusion recall at low decoded precision; strict exact-topology recovery zero for all methods; a calibrated threshold rule dominating edge F1). (4) **Preregistered confirmatory decoder-repair evidence.** A hash-locked one-shot holdout on untouched seeds across three bridges confirms the repair (macro edge F1  $\Delta = +0.0754$ , 95% CI [0.0706, 0.0803]; independent verifier), and a data-only independent temporal audit (Section IV-G) confirms real flow-level fan-out existence outside the development corpus while making no external-performance claim.

On a development-sealed holdout, RC-UOT-Q also improves the precision/F1 frontier over heuristic and style-adapted diagnostic baselines (Table VIII)—not original-system Connector or ABCTracer runs; original-system Connector behavior is reported separately (Section IV-C1; Supplement Appendix A).

## II. RELATED WORK

**Blockchain AML and tracing.** Single-chain AML work uses address clustering, heuristics, and graph propagation [10]–[13], econometric quantification of illicit use [14], and graph-learning pipelines—Elliptic/Elliptic2, dynamic-GNN and inductive representation methods, TRacer-style scalable account-graph tracing, and phishing/scam classifiers [6], [15]–[19]—plus related surveys [20], [21]. Industry reports quantify cross-chain laundering scale but provide no open flow-level correspondence labels [1], [2]. CSFFC shifts the unit of analysis from isolated transactions to cross-chain flows with explicit fan-out/merge and partial observability.

**Cross-chain tracing and bridges.** Bridges create observability seams across ledgers; systematic studies characterize bridge attack surfaces [22], and security research documents DeFi attacks, systemic risks, and attack surfaces [23]–[25]. CONNECTOR [4] mines bridge contract traces to associate deposit/withdrawal pairs, and ABCTracer [5] adds bidirectional tracing and bridge-agnostic cues; both emit one-to-one transaction associations, and we found no published cross-chain tracing system with a native flow-level non-one-to-one correspondence output (audit coverage recorded in the reproducibility package). Earlier CeFi-bridge approaches rely on centralized APIs [26]; our benchmark is anchored

on Celer/cBridge traffic [27]. In Table VIII, CONNECTOR and ABCTracer appear as heuristic/style-adapted diagnostic baselines—not original-system runs; original-system Connector behavior is reported separately (Section IV-C1; Supplement Appendix B).

**Optimal transport and conditional matching prior art.** OT provides principled soft assignment [9], [28]; entropic Sinkhorn scaling [7], unbalanced OT [8], [29], and iterative scaling schemes [30] make it scalable, and OT losses serve distribution matching [31], [32]. The ingredients of our decoder are standard and not claimed as novel: the factorization  $P = \text{diag}(u)K\text{diag}(v)$  is textbook entropic-OT structure [7], [28], and bidirectional score normalization with mutual top- $k$  selection belongs to the dual-softmax/mutual-nearest-neighbor family [33], [34], with Sinkhorn transport plus mutual selection as in SuperGlue [35]. Our contribution at this level is the diagnosis, the cancellation analysis (Section III-C), and its preregistered confirmatory validation; these works are cited as lineage context, not formula-level attributions.

**Explainability and deployment.** Regulation requires risk-based AML/CFT controls [3], [36], and high-stakes outputs should be interpretable and auditable [37]–[39]; reproducibility norms motivate frozen benchmarks and open evaluation scripts [40], [41]. Our deployment boundary is investigator triage—not autonomous adjudication or legal proof. Relative to transaction-level tracers, CSFFC/RC-UOT-Q operate on AML flows, use soft transport mass for fan-out/merge, represent unmatched observation explicitly, and surface evidence gaps through coverage tiers and abstention; we do not claim universal dominance over original Connector, original ABCTracer, or their diagnostic adaptations.

### III. METHOD: EVIDENCE-AWARE FLOW CORRESPONDENCE WITH RC-UOT-Q

#### A. Design Goals and Flow Construction

Cross-chain AML correspondence differs from single-chain tracing in four ways that drive the design: **flow-level modeling** (flows are the observation unit; transaction provenance is retained), **soft correspondence** (a nonnegative transport plan between flow sets, because hard one-to-one matching cannot represent partial mass allocation), **evidence qualification** (hypotheses are conditioned on actually observed events, logs, and receipts), and **abstention and auditability** (unsupported links are withheld rather than promoted) [3], [37]. Outputs are investigation leads for human review, not legal conclusions.

A *flow* aggregates bridge-relevant activity by primary address, chain, asset context, and a rolling 1800-s window, and is written as

$$s_i = (A_i, c_i, g_i, \tau_i, x_i, E_i), \quad (1)$$

with primary address set  $A_i$ , chain  $c_i$ , asset/route context  $g_i$ , temporal summaries  $\tau_i$ , feature vector  $x_i$ , and evidence multiset  $E_i$  (supporting transactions, logs, parsed bridge events); destination flows  $t_j$  are analogous. Evidence collection uses standard JSON-RPC (and optional trace tiers).

#### B. RC-UOT: Risk-Constrained Unbalanced Flow Matching

Given source set  $\mathcal{S} = \{s_1, \dots, s_n\}$  and destination set  $\mathcal{T} = \{t_1, \dots, t_m\}$ , RC-UOT seeks a plan  $P \in \mathbb{R}_{\geq 0}^{n \times m}$  where  $P_{ij}$  allocates correspondence mass from  $s_i$  to  $t_j$ . The pairwise cost combines amount mismatch, temporal causality and delay feasibility, bridge/route consistency, AML risk alignment, and evidence quality:

$$C_{ij} = w_{\text{amt}} c_{\text{amt}}(i, j) + w_{\text{time}} c_{\text{time}}(i, j) + w_{\text{route}} c_{\text{route}}(i, j) + w_{\text{risk}} c_{\text{risk}}(i, j) + w_{\text{evid}} c_{\text{evid}}(i, j). \quad (2)$$

With nonnegative marginal mass vectors  $\mu, \nu$  (whose totals need not be equal), RC-UOT solves

$$\min_{P \geq 0} \langle C, P \rangle + \varepsilon \Omega(P) + \lambda_s \text{KL}(P \mathbf{1}_m \| \mu) + \lambda_t \text{KL}(P^\top \mathbf{1}_n \| \nu), \quad (3)$$

where  $\Omega(P) = \sum_{ij} P_{ij}(\log P_{ij} - 1)$  is entropic regularization [7] and the KL penalties relax exact marginal matching [8], [29], yielding *unmatched mass* appropriate for partial observation and decoys [9], [28]. Solutions use entropy-regularized iterative scaling; decoding maps mass to reported edges under the threshold and sparsity rules of Section IV. Stress-test ablations indicate that temporal causality and unbalanced mass are the most sensitive isolated terms; the risk term’s isolated contribution is not separately quantified (“risk-constrained” refers to the frozen risk-aware cost and marginal design).

#### C. Conditional Transport-Plan Decoding: Dual-Cancellation Proposition

**The raw-plan decoding problem.** An entropic plan factorizes as

$$P = \text{diag}(u) K \text{diag}(v), \quad K_{ij} = \exp(-C_{ij}/\varepsilon). \quad (4)$$

Direct decoding of  $P$  (e.g., mutual top- $k$  by row/column ranks of  $P$ ) ranks row  $i$  by  $u_i K_{ij} v_j$ ; the destination-side dual scaling  $v_j$  enters the row ranking multiplicatively (and symmetrically  $u_i$  for columns). These scalings are solver artifacts of marginal competition, not pairwise affinity, so raw-plan ranking can drop kernel-strong edges.

**Directional conditional scores.** With realized masses  $r_i = \sum_j P_{ij}$  and  $c_j = \sum_i P_{ij}$ , define  $S_{ij}^{\text{row}} = P_{ij}/c_j$  ( $c_j > 0$ ) and  $S_{ij}^{\text{col}} = P_{ij}/r_i$  ( $r_i > 0$ ); zero-conditioning-mass cells carry no support and rank last. The decoder emits edge  $(i, j)$  iff both directional ranks are at most  $k$  (mutual top- $k$ ; frozen  $k = 5$ ).

**Proposition 1** (Directional dual cancellation). *Let  $P = \text{diag}(u) K \text{diag}(v)$  with  $u_i > 0, v_j > 0, K_{ij} > 0$ . Then for every cell,  $S_{ij}^{\text{row}} = u_i K_{ij} / \sum_l u_l K_{lj}$  (independent of  $v$ ), and symmetrically  $S_{ij}^{\text{col}} = K_{ij} v_j / \sum_l K_{il} v_l$  (independent of  $u$ ).*

*Proof sketch.*  $c_j = \sum_l P_{lj} = v_j \sum_l u_l K_{lj}$ ; division cancels  $v_j$ . The column case is symmetric ( $r_i = u_i \sum_l K_{il} v_l$ ).  $\square$

**What the cancellation means.** Within a row the ranking equals that of  $K_{ij}/D_j$  with  $D_j = \sum_l u_l K_{lj}$ ; the conditioned side’s common scaling factor is removed *exactly*, while opposite-side competition  $D_j$  remains. The decoder is therefore **dual-cancelled, not kernel-restored**: it does not

TABLE I  
RESEARCH QUESTIONS.

| ID  | Research question                                                                                            |
|-----|--------------------------------------------------------------------------------------------------------------|
| RQ1 | Can the Celer signals support a reproducible CSFFC flow benchmark?                                           |
| RQ2 | Can RC-UOT express fan-out/merge structures under controlled stress?                                         |
| RQ3 | Can RC-UOT-Q provide reliable inference on event-backed covered quotient pairs?                              |
| RQ4 | Does RC-UOT-Q improve precision/F1 over adapted diagnostic baselines (Table VIII; not original-system runs)? |
| RQ5 | Can the Cross AML prototype execute the evidence-to-report pipeline?                                         |

recover  $K$  exactly and retains opposite-side competition; the scores are gauge-invariant. The identity holds for every plan of the factorized form (balanced entropic OT and entropic unbalanced OT with KL penalties alike) [7], [8], [28], [31], so it is **not UOT-specific and not a new normalization**—it is a direct corollary of standard Sinkhorn structure stated for completeness; whether it improves correspondence ranking is an empirical question answered by the preregistered confirmatory holdout (Section IV-B4). Full derivation, strict-positivity handling, and per-cell cancellation verification are in Supplement Appendix G.

#### D. RC-UOT-Q: Quotient Groups, Coverage Qualification, and Abstention

When exact flow-pair attribution is underdetermined (similar amount–time profiles, shared bridge events, missing event keys), promoting a single  $(s_i, t_j)$  risks over-attribution. RC-UOT-Q groups candidates that share bridge-event incidence into quotient groups

$$q = ([S_q], [T_q], E_q), \quad (5)$$

with source/destination multisets  $[S_q], [T_q]$  collapsed under a shared bridge-event key and qualifying evidence  $E_q$ . **Coverage tiers** then qualify output: A/B (event-backed, sufficient observability) may enter high-confidence output; C is diagnostic only; Uncovered *abstains*. Abstention is an AML safety feature: withholding beats inflating recall with unsupported links [3]. Reports expose the correspondence, tier-qualified confidence, evidence used/missing, abstention reasons, and investigator actions, with an explicit claim boundary (human review required; no legal proof or full-population precision/recall claims). The Cross AML prototype (Section I) implements the full evidence-to-report pipeline in dry-run and live-RPC modes.

## IV. EXPERIMENTS

We evaluate CSFFC, RC-UOT, RC-UOT-Q, and the Cross AML prototype along five research questions (RQ1–RQ5, Table I). Metrics are reported on fixed benchmark splits and held-out test partitions not used for model or threshold selection unless noted.

**Settings.** Real Celer supervision supports benchmark construction (RQ1); semi-synthetic structural stress isolates fan-out/merge expression (RQ2); full-anchor admissible decoding

TABLE II  
CELER-SUPERVISED CSFFC BENCHMARK STATISTICS.

| Quantity                                      | Value                              |
|-----------------------------------------------|------------------------------------|
| Celer anchor transaction pairs                | <b>7,296</b>                       |
| ETH source flows / BNB destination flows      | <b>5,735 / 7,122</b>               |
| Supervised flow label rows                    | <b>7,128</b>                       |
| One-to-one / fan-out / merge (edge share)     | <b>72.32% / 27.51% / 0.17%</b>     |
| Segmentation robustness (600/3600 vs. 1800-s) | Flow-label change $\leq \pm 1.1\%$ |

TABLE III  
EDGE-INCLUSION RECALL UNDER SEMI-SYNTHETIC FAN-OUT/MERGE STRESS (FROZEN REFERENCE; NOT EXACT TOPOLOGY RECOVERY).

| Metric                        | Mean         | 95% CI         |
|-------------------------------|--------------|----------------|
| Fan-out edge-inclusion recall | <b>0.946</b> | [0.907, 0.985] |
| Merge edge-inclusion recall   | <b>0.967</b> | [0.913, 1.000] |
| Per-source Recall@3           | 0.481        | [0.418, 0.544] |

on the frozen plan evaluates ranked correspondence without re-solving UOT (Section IV-C); covered-quotient holdout evaluation tests RC-UOT-Q (RQ3); an independent held-out test set compares RC-UOT-Q against adapted diagnostic baselines (RQ4); prototype validation confirms end-to-end execution (RQ5). An independent temporal data audit (Section IV-G; data-only) additionally documents real fan-out structure outside the development corpus—problem-validity evidence, not performance-validation evidence.

#### A. Celer-Supervised Flow Benchmark

Labels derive from official **Celer cross-chain anchor pairs** (one Ethereum hash linked to one BNB Smart Chain hash; no heuristic relabeling). Flows aggregate activity by primary address, chain, asset context, and a rolling **1800-s** window (Table II).

*Topology note: legacy label fields are inverted—legacy many\_to\_one is  $1 \rightarrow N$  fan-out and legacy one\_to\_many is  $N \rightarrow 1$  merge (verified in code and data); numbers are unchanged.*

#### B. Structural Expression of Non-One-to-One Bridge Flows

We stress-test RC-UOT on semi-synthetic episodes with known fan-out, merge, unmatched, and decoy structure cloned from real subgraphs: **48** templates  $\times$  **5** seeds (42–46).

*The fan-out/merge metric counts a ground-truth edge as recovered iff the decoded plan assigns it positive mass ( $\geq 1e-9$ ); edge-inclusion recall over a dense plan, NOT exact topology recovery (Section IV-B2). Recall@3 is the macro-averaged fraction of source flows—with at least one GT destination and one decoded edge  $> 1e-9$ —whose GT destination appears among the top-3 destinations by decoded mass, averaged over seeds 42–46 (rank-based inclusion; per-source macro; full definition in Supplement Appendix C).*

The same frozen faithful run across three bridges gives fan-out/merge edge-inclusion recall 0.950/0.967 (Celer),

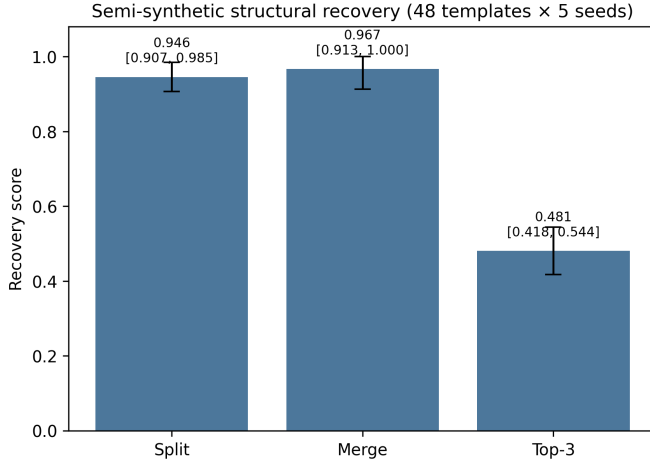


Fig. 3. Structural capability under semi-synthetic fan-out/merge stress: mean edge-inclusion recall with 95% CIs over 48 templates and 5 seeds (inclusion in the dense decoded plan, not exact topology recovery; Table V).

TABLE IV

METHOD CAPABILITY TABLE (CODE-LEVEL AUDIT + CONSTRUCTED UNIT TESTS).

| Method          | Fan-out (1→2) | Merge (2→1) | Global alloc. | Unmatched |
|-----------------|---------------|-------------|---------------|-----------|
| Connector-style | No            | Partial     | No            | Partial   |
| ABCTracer-style | No            | Partial     | No            | Partial   |
| Threshold-MM    | Yes           | Yes         | No            | Yes       |
| Balanced-OT     | Yes           | Yes         | Yes           | No        |
| RC-UOT-Q        | Yes           | Yes         | Yes           | Yes       |

0.971/0.979 (Multichain), 1.000/1.000 (PolyNetwork) for RC-UOT-Q, and 0.000 for both one-to-one baselines on every bridge (Supplement Table S2).

1) *Representation Capability*: Code-level capability tests (TEST A–F) show the structural difference before any fitting: Connector-style / ABCTracer-style decoders enforce source out-degree  $\leq 1$ , so a 1→2 fan-out is structurally unrepresentable (merge is representable in their data structure but their pairwise selection scores zero merge edges here); Threshold-MM decides per cell independently; Balanced-OT emits both edges of fan-out/merge but cannot abstain; RC-UOT-Q relaxes marginals (mass destruction/creation is the first-class abstention signal). Table IV summarizes; the capability ladder figure is in Supplement Fig. S1.

“Unmatched = Partial” for one-to-one methods means rejection-level abstention only, never a first-class per-node decision. RC-UOT-Q’s “Yes” is a mass-level property.

2) *Three-Bridge Structural Expression (Strict Evaluation)*: On identical grids (seeds 42–46), five methods share one unified strict evaluator: strict exact recovery (predicted edge set must equal the structural ground truth exactly; one extra edge forfeits the template), edge precision/recall/F1, degree accuracy, and FP counts. Threshold-MM uses one global threshold calibrated on disjoint seeds 101–103 by edge F1 (frozen cutoff 0.478, never re-tuned on test seeds—an author-defined calibration protocol); Balanced-OT runs with pre-evaluation frozen defaults (unit-normalized marginals, reg =

0.05 equal to the frozen UOT reg, same 1e-9 decode, no labels); the one-to-one baselines use author-set fixed constants with zero calibration budget (Section IV-E).

Source: *structural\_baseline\_mechanism\_study/aggrega* (independent verification recomputes every cell). The 0.946–1.000 of Tables III/S2 is edge-inclusion recall; the strict exact column here is 0.000 for every method on every bridge.

**Interpretation.** Only the many-match methods can emit a 1→2 edge set at all, but at the frozen operating point no method achieves exact topology recovery, and on edge-level metrics the globally calibrated per-cell threshold rule (F1 0.120–0.168) *dominates* both transport methods (F1  $\leq 0.040$ ); RC-UOT-Q’s unbalanced relaxation keeps it above strictly balanced OT on every bridge (precision 0.015–0.022 vs. 0.010–0.016) while its frozen dense decode keeps edge precision low. The all-zero strict column has three audited causes (dense 1e-9 decodes—18,776 edges on the Celer seed-42 grid vs. a 6-edge truth; true structural edges are not locally cheapest, so any local threshold keeps cheaper confusers; and the legacy metric counts inclusion), detailed in Supplement Appendix H. Strict-evaluation and mechanism-ladder figures are in Supplement Figs. S2–S3.

3) *Why Unbalanced Transport (Mechanism Ladders)*: Four seed-controlled ladders (seeds 101–103, identical cost matrices, no re-tuning) isolate the unbalanced mechanism. Clean conditions: at 0% mass mismatch, Balanced-OT is not close to RC-UOT-Q (edge F1 0.070 vs. 0.110 Celer; 0.032 vs. 0.047 Multichain; 0.033 vs. 0.045 PolyNetwork)—shedding mass on confusers matters even at the balanced point. Under mass mismatch both transport methods degrade (Celer precision 0.043→0.024 balanced; 0.068→0.037 RC-UOT-Q) while Threshold-MM is flat or improves (0.142→0.155); RC-UOT-Q stays above Balanced-OT at every level as a level offset, not a slope difference. Under unmatched ratios, all methods lose precision while RC-UOT-Q destroys the excess mass (its transported mass falls—the explicit abstention signal). Under decoy density, Threshold-MM’s pairwise-independent false positives grow fastest (26.3→223.6 FP per template at  $k=8$ , +750%; precision 0.182→0.061) vs. slower joint-method growth (+278%/+239%); timestamp noise ( $\leq 180$ -s) does not discriminate the methods. The data therefore support the weakest preregistered conclusion: RC-UOT-Q removes the representational limitation of one-to-one hard matching; they do not support global-transport consistency superiority over the calibrated local rule, and unbalanced mass robustness is supported only as a consistent edge-metric level offset, not exact-topology robustness (full ladder numbers in Supplement Table S6).

4) *Confirmatory Holdout: Conditional-Plan Decoding Repairs Raw-Plan Correspondence Ranking*: On development seeds 201–205, amount-free cost decoding attained macro edge F1 0.3172 while raw-plan mutual-top-5 decoding (RAW\_UOT\_PLAN\_D4) attained only 0.2374; the diagnosis traced the gap to marginal competition and destination-side dual scaling. We preregistered a dual-cancelled conditional decoder ( $S^{\text{row}} = P_{ij}/c_j$ ,  $S^{\text{col}} = P_{ij}/r_i$ , mutual top-5,  $k=5$ ) with a five-gate decision tree before any holdout data existed. All components—amount-free cost, risk/evidence-weighted

TABLE V

STRICT STRUCTURAL EVALUATION, THREE BRIDGES  $\times$  FIVE METHODS (48 TEMPLATES  $\times$  5 SEEDS; PER-SEED = MEAN OVER TEMPLATES; MEAN  $\pm$  S.D. OVER SEEDS).

| Bridge      | Method            | Fan-out exact | Merge exact | Edge P            | Edge R            | Edge F1           | Deg. acc          | FP edges           |
|-------------|-------------------|---------------|-------------|-------------------|-------------------|-------------------|-------------------|--------------------|
| Celer       | Connector-style   | 0.000         | 0.000       | 0.026 $\pm$ 0.007 | 0.026 $\pm$ 0.007 | 0.026 $\pm$ 0.007 | 0.000             | 5.84 $\pm$ 0.04    |
| Celer       | ABCTracer-style   | 0.000         | 0.000       | 0.026 $\pm$ 0.007 | 0.026 $\pm$ 0.007 | 0.026 $\pm$ 0.007 | 0.000             | 5.84 $\pm$ 0.04    |
| Celer       | Threshold-MM      | 0.000         | 0.000       | 0.094 $\pm$ 0.009 | 0.971 $\pm$ 0.006 | 0.168 $\pm$ 0.016 | 0.004 $\pm$ 0.064 | 67.7 $\pm$ 9.7     |
| Celer       | Balanced-OT       | 0.000         | 0.000       | 0.016 $\pm$ 0.002 | 0.968 $\pm$ 0.036 | 0.030 $\pm$ 0.005 | 0.002 $\pm$ 0.046 | 680.8 $\pm$ 128.9  |
| Celer       | RC-UOT-Q (frozen) | 0.000         | 0.000       | 0.022 $\pm$ 0.005 | 0.961 $\pm$ 0.035 | 0.040 $\pm$ 0.009 | 0.002 $\pm$ 0.046 | 539.0 $\pm$ 135.3  |
| Multichain  | Connector-style   | 0.000         | 0.000       | 0.000             | 0.000             | 0.000             | 0.000             | 6.00               |
| Multichain  | ABCTracer-style   | 0.000         | 0.000       | 0.034 $\pm$ 0.006 | 0.034 $\pm$ 0.006 | 0.034 $\pm$ 0.006 | 0.000             | 5.80 $\pm$ 0.04    |
| Multichain  | Threshold-MM      | 0.000         | 0.000       | 0.075 $\pm$ 0.005 | 0.844 $\pm$ 0.030 | 0.136 $\pm$ 0.009 | 0.000             | 71.0 $\pm$ 2.9     |
| Multichain  | Balanced-OT       | 0.000         | 0.000       | 0.010 $\pm$ 0.003 | 0.994 $\pm$ 0.008 | 0.018 $\pm$ 0.005 | 0.000             | 1069.6 $\pm$ 142.4 |
| Multichain  | RC-UOT-Q (frozen) | 0.000         | 0.000       | 0.015 $\pm$ 0.004 | 0.983 $\pm$ 0.020 | 0.027 $\pm$ 0.008 | 0.000             | 925.4 $\pm$ 140.7  |
| PolyNetwork | Connector-style   | 0.000         | 0.000       | 0.000             | 0.000             | 0.000             | 0.000             | 6.00               |
| PolyNetwork | ABCTracer-style   | 0.000         | 0.000       | 0.000             | 0.000             | 0.000             | 0.000             | 6.00               |
| PolyNetwork | Threshold-MM      | 0.000         | 0.000       | 0.065 $\pm$ 0.006 | 0.997 $\pm$ 0.006 | 0.120 $\pm$ 0.010 | 0.000             | 107.5 $\pm$ 8.1    |
| PolyNetwork | Balanced-OT       | 0.000         | 0.000       | 0.013 $\pm$ 0.001 | 1.000             | 0.026 $\pm$ 0.002 | 0.000             | 852.8 $\pm$ 55.4   |
| PolyNetwork | RC-UOT-Q (frozen) | 0.000         | 0.000       | 0.018 $\pm$ 0.004 | 1.000             | 0.034 $\pm$ 0.007 | 0.000             | 743.6 $\pm$ 63.4   |

marginals,  $\text{reg} = 0.05$ ,  $\text{reg\_m} = 0.5$ , decoder, paired bootstrap ( $B = 4,000$ ,  $\text{RandomState}(20240101)$ ), decision rules—were hash-locked before the untouched seeds 301–305 (three bridges, 48 templates per seed, 720 paired template instances) were generated or read; the one-shot runner verifies every hash, and an independent verification pass recomputes all results from raw per-cell artifacts within  $1e-9$ .

**Primary result.** RAW\_UOT\_PLAN\_D4 attains macro edge F1 0.2350 and the conditional decoder 0.3104:  $\Delta_{\text{primary}} = +0.075413$ , **paired 95% CI** [0.070557, 0.080312], entirely above zero (gate A; Table VI; Fig. 4). The CI is a within-bridge template-level paired resample (the seed level enters the point estimate but not the resampling); per-bridge paired s.d. 0.0704/0.0881/0.0247. The improvement is positive on every bridge—Celer  $+0.0826$  [0.0743, 0.0918], Multichain  $+0.0820$  [0.0709, 0.0933], PolyNetwork  $+0.0616$  [0.0587, 0.0649] (gate D)—so the primary effect is not driven by a single bridge.

**Mechanism gate (B).** The exact preregistered directional signature holds: harmful-flip rates decrease by  $-0.2595$  (row),  $-0.2303$  (column),  $-0.5360$  (fan-out child),  $-0.5095$  (merge destination), and GT mutual-top5 retention increases by  $+0.2035$ . These observational indicators are consistent with the ranking-repair mechanism, not a complete solver-level causal decomposition. **Recovery (gate C).** With  $D = +0.070569$ ,  $\text{recovery\_fraction} = 1.0686$  (point estimate; the amount-free cost reference was a reference, not a ceiling).

**Transport-plan contribution and UOT/BOT.** Conditional decoding of the transport plan exceeds support-plus-kernel decoding ( $\Delta_{\text{support}} = +0.0104$  [0.0062, 0.0152]) and amount-free cost decoding ( $\Delta_{\text{cost}} = +0.0048$  [0.0013, 0.0089])—positive transport-plan-level contribution at the macro level (TYPE A; PolyNetwork’s secondary contrasts are negative:  $\Delta_{\text{cost}} = -0.0043$  [−0.0065, −0.0024],  $\Delta_{\text{support}} = -0.0043$  [−0.0065, −0.0024], so TYPE A is a three-bridge macro classification, not bridge-uniform). This is not a claim about the unbalanced relaxation: CONDITIONAL\_BOT\_D4 (0.3106) and CONDITIONAL\_UOT\_D4 (0.3104) differ by  $\Delta_{\text{bot}} \approx -0.0003$  (unrounded macro mean  $-0.0002688$ ; 95% CI [−0.0018, +0.0012], includes zero), so **no unbalanced-relaxation-specific benefit is es-**

TABLE VI  
PREREGISTERED CONFIRMATORY HOLDOUT (SEEDS 301–305), FIVE  
FROZEN DECODING METHODS.

| Method              | Edge precision (macro) | Edge recall (macro) | Edge F1 (macro) |
|---------------------|------------------------|---------------------|-----------------|
| RAW_UOT_PLAN_D4     | 0.1476                 | 0.5975              | 0.2350          |
| CONDITIONAL_UOT_D4  | 0.1926                 | 0.8009              | 0.3104          |
| AMOUNT_FREE_COST_D4 | 0.1896                 | 0.7882              | 0.3055          |
| CONDITIONAL_BOT_D4  | 0.1927                 | 0.8016              | 0.3106          |
| SUPPORT_PLUS_K_D4   | 0.1862                 | 0.7734              | 0.2999          |

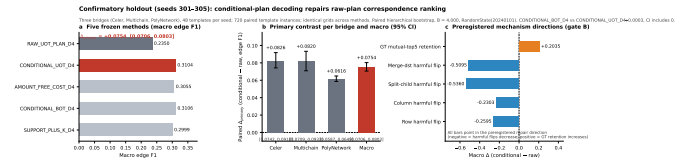


Fig. 4. Preregistered confirmatory holdout (seeds 301–305; 720 paired template instances). (a) Macro edge F1 of the five frozen decoding methods; primary comparison CONDITIONAL\_UOT\_D4 vs. RAW\_UOT\_PLAN\_D4 ( $\Delta_{\text{primary}} = +0.0754$  [0.0706, 0.0803]). (b) Paired  $\Delta$  per bridge and macro (all positive). (c) Five preregistered mechanism-direction changes: all four harmful-flip rates decrease and GT mutual-top5 retention increases. CONDITIONAL\_BOT\_D4 and CONDITIONAL\_UOT\_D4 differ by  $-0.0003$  with a CI including 0; no UOT-over-BOT implication.

**tablished.** Integrity: FP/template 1.0511, FN/template 0.4945 (within the  $3 \times$  bound), max solver residuals  $\leq 10^{-11}$ , zero NaNs/failures/duplicates.

**Limitations.** The endpoint is template-level edge F1 on semi-synthetic structure (strict exact-topology recovery remains zero; Table V);  $k = 5$ ; mechanism indicators are observational.

*Primary contrast:* **CONDITIONAL\_UOT\_D4** – **RAW\_UOT\_PLAN\_D4** (**bold rows**);  $\Delta_{\text{primary}} = +0.075413$ , **CI** [0.070557, 0.080312] ( $B = 4,000$ ,  $\text{RandomState}(20240101)$ ). **CONDITIONAL\_BOT\_D4** vs. **CONDITIONAL\_UOT\_D4**:  $-0.0003$ , **CI includes 0**. *Values transcribed from the frozen consolidation records; raw per-seed artifacts were not accessed.*

TABLE VII  
FIXED-DELAY RC-UOT-Q MAIN RESULTS (FULL ANCHOR PAIRS;  
DECODE-ONLY; TRANSPORT FROZEN).

| Decoding                                | Pair prec. | Pair rec. | Pair F1 | Top-3 rec. | tx-CVR | Cov.  | Abst. |
|-----------------------------------------|------------|-----------|---------|------------|--------|-------|-------|
| Raw tx projection (compatibility)       | 0.589      | 0.589     | 0.589   | 0.658      | 0.338  | 1.000 | 0.000 |
| Top-3 admissible rescue (high coverage) | 0.590      | 0.589     | 0.590   | 0.658      | 0.005  | 0.996 | 0.002 |
| Joint time-admissible filter (forensic) | 0.889      | 0.589     | 0.708   | 0.658      | 0.000  | 0.845 | 0.338 |
| Permuted-label control                  | 0.000      | 0.000     | 0.000   | 0.002      | —      | —     | —     |

### C. Fixed-Delay RC-UOT-Q on Full Anchor Pairs

Using the `tx_if_available_else_flow_representative` delay policy, RC-UOT is evaluated on all **7,296** Celer anchor pairs as a ranked flow-correspondence model; the frozen plan is *not re-solved*—admissibility applies decode-only temporal masks and top- $k$  rescue (Table VII). The raw tx-level argmax projection (pair F1 0.589, top-3 recall 0.658) is a *compatibility projection* only (tx-level CVR 0.338), not a forensic admissibility claim. The formal high-coverage decoding (`positive_delay_top3_rescue`) reaches pair F1 0.590 with coverage 0.996 and tx-CVR 0.005; the forensic high-confidence subset (`joint_time_admissible_filter`) yields precision **0.889** and pair F1 **0.708** at coverage **0.845** with tx-CVR **0**. A permuted-label control collapses pair P/R/F1 to near zero. Coverage (source flows with a non-abstained projection) and abstention (all 7,296 pairs) denominators differ by design. The fixed-delay leave-anchor-out audit shows `forbidden_features_remaining` = 0 with fake-anchor probes neutralized, and baseline/leave-key-out/strict agree within 0.02—the gains are not explained by bridge-key or bridge-evidence leakage (Supplement Table S7).

Coverage and abstention denominators differ by design. Permuted-label control cells “—” are not method performance. Legacy pre-admissible metrics (pair F1  $\approx 0.321$ ) are diagnostic-only.

1) *Symmetric Masking and Baseline Applicability*: Under a symmetric masking ladder on the shared 7,296 tx-pair ground truth and BNB candidate universe (Supplement Table S3), original Connector (`WithdrawLocator` raw top-1) reaches F1 **0.9736** under full native bridge semantics and the closed-set candidate pool—an upper-bound native-feature diagnostic, since the pool equals the 7,296 GT destination hashes; RC-UOT-Q (joint filter) reaches **0.7085**. Removing receiver or amount semantics blocks or collapses Connector (receiver: `BLOCKED_BY_REQUIRED_RECEIVER_SEMANTICS`, N/A; amount: zero predictions, F1 0.0000), while RC-UOT-Q degrades (0.7001/0.3714/0.3752) but remains evaluable with tx-CVR = 0 throughout. **The contribution is not that RC-UOT-Q dominates Connector in native mode**: RC-UOT-Q provides a ranked flow-correspondence model with an admissible abstaining decoder that stays applicable when bridge-specific matching semantics are absent. Route A v1 (inflated F1 0.9953) was rejected for decimal-bootstrap inconsistency; ABCTracer remains blocked (no official checkpoint).

TABLE VIII  
FLOW-STRESS COMPARISON ON THE INDEPENDENT HELD-OUT TEST SET  
(CALIBRATION ASYMMETRY DISCLOSED IN THE TEXT).

| Method                                 | Precision    | Recall | F1           | ECE   |
|----------------------------------------|--------------|--------|--------------|-------|
| <b>RC-UOT-Q</b>                        | <b>0.192</b> | 0.900  | <b>0.316</b> | 0.028 |
| Heuristic adapted (Connector-inspired) | 0.113        | 0.718  | 0.195        | 0.060 |
| Style-adapted (ABCTracer-inspired)     | 0.100        | 0.975  | 0.181        | 0.018 |

### D. Coverage-Qualified Quotient Inference

On the **122** covered holdout pairs (44/78; seeds 212–231), the development-frozen bridge-transfer-key rule reproduces the transferId-consistent quotient grouping exactly (P/R/F1 = 1.000; Supplement Table S1). This is a **definitional consistency check** on the same protocol-native transferId association rule—not independent-accuracy and not external-performance evidence. Event-backed projection coverage is  $\approx 0.792$  (gate-reference seeds 52–57; the full-scope gate fails on this value), so covered metrics must not be read as full-population recovery.

### E. Comparison with Cross-Chain Tracing Baselines

RC-UOT-Q (precision-oriented reranking, development-frozen) is compared with heuristic/style-adapted diagnostic baselines—not original-system runs—on an independent development-sealed flow-stress holdout (Table VIII). **Calibration asymmetry disclosed**: RC-UOT-Q received a development tuning budget (logistic precision reranker trained on dev seeds 52–71; threshold  $\tau = 0.7796$  selected on dev; decoder selected on dev; all frozen before holdout 292–311 was touched), whereas the two adapted baselines have zero calibration budget (author-set fixed constants in the frozen scripts/config catalog, `calibration="none"`, threshold 0.0, top-k 50, no development-selection step; the repository contains no author statement that these constants were never implicitly adjusted, so their historical tuning status is stated at its most conservative level). The budget is **asymmetric and favors the proposed method**; Table VIII is therefore a fixed-scope diagnostic with untuned heuristic controls—diagnostic/structural/native-semantic controls, not a budget-matched leaderboard, and not evidence of calibrated superiority. RC-UOT-Q improves precision and F1 (0.192/0.316) over both baselines (0.113/0.195; 0.100/0.181) while the ABCTracer-inspired baseline retains higher recall (0.975 vs. 0.900); these are single-partition point estimates without CIs (exploratory, RQ4). Original-system comparison is in Section IV-C1/Supplement Appendix A. On the cached Multi/Poly extension, adapted baselines remain strong on near one-to-one anchors (Multichain 0.978/0.959 vs. RC-UOT-Q 0.010 with 0.984 abstention; PolyNetwork 0.998/0.998 vs. RC-UOT-Q F1 0.833 at coverage 0.881)—near-total abstention is the designed fail-safe of a coverage-qualified model (Supplement Table S4).

*Not original-system runs (Route A v2 and the Connector native audit supersede for original-baseline claims; native Connector F1 = 0.9736). Point estimates, no CI; exploratory.*

### F. Prototype Validation and Summary

**Cross AML** executes the evidence-to-report pipeline end-to-end in dry-run mode (replaying bundled evidence without RPC credentials): input parsing, evidence replay, flow construction, coverage qualification, RC-UOT-Q-style scoring, and report generation (RQ5). The experiments support six conclusions: **(1) Benchmark**—the Celer corpus provides a reproducible flow-level CSFFC setting (7,296 pairs; 7,128 labels). **(2) Structural transport**—RC-UOT expresses fan-out/merge in the edge-inclusion sense (0.946/0.967); strict exact-topology recovery is 0.000 for every method, and the calibrated threshold rule dominates edge F1 (Table V). **(3) Decoder repair**—the preregistered holdout confirms conditional decoding repairs raw-plan ranking (+0.0754 [0.0706, 0.0803], positive on every bridge, all mechanism directions satisfied). **(4) Full-anchor admissible decoding**—top-3 rescue preserves coverage 0.996 with tx-CVR 0.005; the joint filter yields precision 0.889/F1 0.708 at coverage 0.845 (Table VII). **(5) Covered quotient and baselines**—the 122-pair consistency check, the masking applicability curve, and the Table VIII precision/F1 improvement with its calibration caveat. **(6) Independent real-data audit**—2,451 Tier-A fan-out units over 359.98 days confirm problem validity outside the development corpus, with no method executed (Section IV-G).

### G. Independent Temporal Data Audit (Problem Validity; No Method Performance)

A disjoint post-development Celer corpus (window 2024-05-20T00:00:00Z–2025-05-19T00:00:00Z; protocol-native linkage Send.transferId == Relay.srcTransferId; per-block continuity, independent GT verification, and disjointness against 359,256 historical transaction identities with zero intersections) was assembled data-only and prediction-blind, yielding **32,905** anchors, **25,621** unique flow-level GT edges, **2,451** Tier-A source-level fan-out units (degree  $\leq 5$ : 2,425; degree  $> 5$ : 26), and **85** merge units over a **359.98-day** span (Table IX). These units concentrated in **seven** independent primary-address clusters (910, 587, 452, 432, 60, 8, 2; max share **0.371277**), below the preregistered adequacy gate  $G \geq 8$ . Accordingly, no method predictions were generated and **no external-performance claim is made**; the audit is problem-validity evidence, and it is a different experiment from the 301–305 holdout.

**Why the gate is what it is.**  $G$  counts independent primary-address clusters—a proxy for *behavioral-source diversity*, not sample count: 2,451 fan-out units concentrated in seven clusters cannot be treated as 2,451 independent external actors. The gate was a **conservative, prespecified adequacy requirement, not a universal statistical threshold**, fixed before the corpus structure or any method outcome was observed (jointly with max share  $< 0.5$ , Tier-A units  $\geq 30$ , span  $\geq 2$  calendar months). Because v4 yielded  $G = 7$ , the criterion was *not* relaxed post hoc—that would be a post-data change to a confirmatory rule—and no method prediction existed when adequacy failed. Protocol-level Celer settlement is transaction-pairwise; non-one-to-one structure arises at the forensic flow

TABLE IX  
INDEPENDENT TEMPORAL DATA AUDIT (DATA-ONLY; NO METHOD EXECUTION).

| Quantity                                     | Value                                     |
|----------------------------------------------|-------------------------------------------|
| Protocol-native anchors                      | <b>32,905</b>                             |
| Unique flow-level GT edges                   | <b>25,621</b> (0 duplicates)              |
| Tier-A source-level fan-out units            | <b>2,451</b>                              |
| Merge units                                  | <b>85</b>                                 |
| Temporal span                                | <b>359.98 days</b>                        |
| Independent primary-address clusters ( $G$ ) | <b>7</b> ([910, 587, 452, 432, 60, 8, 2]) |
| Maximum cluster share                        | <b>0.371277</b>                           |
| Preregistered adequacy gate ( $G \geq 8$ )   | <b>FAIL</b>                               |
| Method predictions / performance             | <b>None</b>                               |

level when settlements aggregate (the long-form justification is in Supplement Appendix E).

*v4 Level-I data-only package; prediction-blind, archived for provenance, not an evaluation population.*

## V. DISCUSSION

### A. What the Results Show

**Flow-level transport.** Semi-synthetic stress shows that risk-constrained unbalanced transport expresses fan-out/merge in the edge-inclusion sense (0.946/0.967; Section IV-B)—inclusion recall at decoded edge precision  $\approx 0.02$ , with strict exact-topology recovery **0.000** for every method on every bridge and the globally calibrated per-cell threshold rule dominating edge F1 (Table V). Soft, mass-bearing correspondence is a viable representational alternative to brittle one-to-one matching, under controlled semi-synthetic conditions only.

**Conditional-plan decoder repair.** The preregistered, hash-locked holdout established that dual-cancelled conditional decoding repairs the raw plan’s correspondence ranking (+0.0754 [0.0706, 0.0803], positive on every bridge, mechanism directions satisfied, independent verification; Table VI; Fig. 4). The same holdout does not attribute the gain to the unbalanced relaxation (balanced vs. unbalanced conditional decoding:  $-0.0003$ , CI includes zero). The transport-plan-level increment over direct kernel decoding is positive but small ( $\Delta_{\text{cost}} = +0.0048$  [0.0013, 0.0089];  $\Delta_{\text{support}} = +0.0104$  [0.0062, 0.0152]); the transport solve is a representation mechanism, not a large ranking gain over its own kernel.

**Full-anchor admissible decoding.** The raw projection is a compatibility projection only (pair F1 0.589, tx-CVR 0.338); high-coverage rescue reaches 0.590 with coverage 0.996 and tx-CVR 0.005, and the forensic high-confidence subset yields precision 0.889 / F1 0.708 at coverage 0.845 (Table VII); gains are not explained by bridge-key or bridge-evidence leakage.

**Covered quotient and baselines.** The 122-pair result is a definitional consistency check (Section IV-D); the Table VIII precision/F1 improvement is specific to the flow-stress operating point—on the cached raw transaction-pair extension the adapted baselines stay stronger, and under native closed-set semantics Connector reaches 0.9736 (Section IV-C1).

**Executable workflow.** The Cross AML prototype runs the evidence-to-report pipeline standalone, including dry-run replay (RQ5).

### B. Coverage Qualification as an AML Safety Mechanism

AML automation should fail safely when evidence is incomplete [3], [37]. Coverage qualification promotes hypotheses only under event-backed quotient structure and parser coverage; otherwise the system abstains or emits diagnostic leads. Event-backed projection coverage is  $\approx 0.792$ ; the 1.000 covered P/R/F1 on 122 pairs is rule consistency within the covered scope (Supplement Table S1), *not* full-universe recovery. Coverage tiers and abstention are a design-time safety mechanism, not a post-hoc excuse for weak global recall.

### C. Precision Trade-offs vs. Recall-Oriented Retrieval

RC-UOT-Q is precision-emphasizing and evidence-qualified; the ABCTracer-inspired adaptation remains useful for recall-oriented candidate retrieval (its higher recall, 0.975, is not a claim about the original system). **Absolute-precision honesty:** at the flow-stress operating point precision is 0.192 (about four of five promoted edges are false positives); at the high-recall structural point decoded precision is  $\approx 0.02$ ; strict exact-topology recovery is 0 for every method; and the calibrated threshold heuristic is stronger on edge F1 (0.120–0.168 vs.  $\leq 0.040$ ). What this paper establishes is therefore **representation capability and decoder ranking repair—not investigator-ready automatic structure reconstruction**. The one healthy-precision figure (0.889) lives in the one-to-one anchor regime where closed-set Connector reaches 0.9736, so no operating point with both investigator-grade precision and non-one-to-one coverage is claimed. The practical implication is complementary deployment: recall-oriented tracing widens the candidate pool; RC-UOT-Q ranks and qualifies flow-level hypotheses for triage.

### D. Observability, Cold Start, and Deployment

Cross-chain correspondence quality depends on what investigators can observe: bridge event schemas differ; standard JSON-RPC may lack historical logs/traces; off-chain flows break on-chain continuity; large-scale candidate discovery needs indexer/archive/trace backends. Cold start is handled by a graded response—high-confidence covered match (A/B), low-confidence lead, abstention (Uncovered/Tier C), or an evidence-gap report—with coverage tier, abstention reasons, and discovery limits surfaced explicitly [38]. Cross AML is a research prototype for triage; outputs require human review before external escalation.

### E. Scope of Claims

Final claims are intentionally scoped. These results do not establish: universal superiority over original Connector, original ABCTracer, or their diagnostic adaptations; full-scope high precision/recall (projection coverage  $\approx 0.792$ ); key-metric dominance across dimensions; balanced dominance over all baselines; **external method performance on a fully adequate independent temporal corpus** (the data-only audit failed the preregistered adequacy gate and no method was executed); or a UOT-specific advantage over balanced OT ( $\Delta_{\text{bot}}$  CI includes zero). Industry threat intelligence [1], [2] motivates the setting but is not evidence of model dominance.

### F. Limitations and Future Work

**External method performance remains unestablished.** An initial preregistered Level-I external execution entered the prediction path and failed before valid performance (implementation defect; zero-marginal solver-domain defect) and was not rerun. The solver-interface repair was mathematically justified and validated only on non-test data before a new disjoint corpus was collected; that corpus failed the preregistered adequacy gate ( $G = 7 < 8$ ), so no method predictions were generated and no external-performance claim is made. The v4 fan-out concentration (seven clusters, max share 0.371277) is a dataset-specific descriptive observation about that window, not a claim about the wider ecosystem. **Protocol settlement vs. flow structure:** protocol-level bridge settlement is transaction-pairwise; non-one-to-one structure arises at the forensic flow level. **UOT-specific advantage:** not established ( $\Delta_{\text{bot}}$  CI includes zero); the unbalanced relaxation is motivated by unmatched mass, partial/open-world correspondence, and abstention-compatible modeling, not by proven ranking superiority. **Real merge evidence:** limited (85 merge units in v4; 0.17% merge share in Table II); semi-synthetic stress remains necessary for controlled merge evaluation. The canonical benchmark is Celer-centered (the Multi/Poly extension is a cached raw tx-pair scope with adapted baselines, not original-system ABCTracer); supervision relies on official anchors; strict exact-topology recovery is zero on this benchmark family;  $k = 5$ ; mechanism indicators are observational. Operational limits include bridge-event observability, parser coverage, limited multi-seed statistical power, and discovery backends. Future work: broader bridge/parser coverage, live deployment with archive/trace tiers, larger seed sweeps with tighter CIs, investigator-in-the-loop studies. Any future external-validation study requires a new independent scientific question and fresh preregistration; no new window may be opened merely to satisfy  $G \geq 8$ .

### G. Ethical Use

Reports provide investigation leads, not adjudicated guilt or legal proof; human review is mandatory before escalation; suppressing abstention to inflate recall undermines both AML practice and the method's intent. Investigators should combine on-chain hypotheses with off-chain intelligence, legal process, and institutional policy.

### H. Experimental Provenance and Reproducibility

The evaluation chain is recorded because it is part of the reproducibility asset: **development**  $\rightarrow$  **frozen candidate** (sha256 0f360addc1f2a220299d75b4aa9cad1e1ee1e504e8097ae5bacfc0da40542cc401)  $\rightarrow$  **301–305 confirmatory holdout**  $\rightarrow$  **v3 external execution failure**  $\rightarrow$  **no rerun**  $\rightarrow$  **support-aware solver-interface repair**  $\rightarrow$  **non-v3 equivalence tests**  $\rightarrow$  **v4 disjoint corpus**  $\rightarrow$  **adequacy fail**  $\rightarrow$  **no method execution**. Two distinct repairs must not be conflated: (i) the *decoder repair* (Proposition 1, confirmed on 301–305); and (ii) the *solver-interface repair* (support-aware wrappers, validated only on non-test data,

never used for any performance number). Every confirmatory number was produced by the hash-locked one-shot runner and recomputed by the independently maintained verifier within  $1e-9$ ; the full chronology, seed map, and frozen hashes are in the Supplement (Appendices D–F).

### I. Real-Data Fan-Out Concentration and Availability

The v4 concentration (seven clusters, max share 0.371277) means fan-out in that window was concentrated among a small number of aggregators or routers; it licenses no generalization that the ecosystem is dominated by seven actors and says nothing about method performance. Code and reproducibility materials will be provided upon acceptance/public release; the release address and license declaration will be supplied in the submission package. The package includes the Cross AML prototype, the frozen scripts (one-shot runner + verifier), hash manifests, frozen confirmatory artifacts, the v3 failure package, and the v4 data-only corpus (prediction-blind, with its dataset hash manifest). Raw chain data are re-fetchable from public Ethereum/BNB Smart Chain nodes; provider keys are not released; Celer label redistribution terms are stated in the package.

## VI. CONCLUSION

Cross-chain laundering forces AML analysis beyond single-chain transaction tracing toward flow-level correspondence across bridges, tokens, and ledgers. We introduced **CSFFC** as the task formulation, with flows as AML observation units supporting fan-out, merge, delay, and partial observability. **RC-UOT** solves CSFFC through risk-constrained unbalanced optimal transport with explicit unmatched mass; semi-synthetic stress achieves fan-out/merge edge-inclusion recall of **0.946/0.967**—inclusion recall at decoded edge precision  $\approx 0.02$ —while strict exact-topology recovery is zero for every tested method and a calibrated threshold rule dominates edge F1 (Table V); a code-level capability audit shows one-to-one decoders structurally cannot represent one-to-many correspondence.

**Conditional decoding.** We diagnosed that raw-plan decoding mixes kernel ranking with solver dual scalings, and a preregistered, hash-locked confirmatory holdout on untouched seeds across three bridges confirms the repair: dual-cancelled conditional decoding raises macro edge F1 by  $+0.0754$  (paired 95% CI [0.0706, 0.0803]), positive on every bridge, with all five preregistered mechanism-direction indicators satisfied and an independent verifier recomputing every result. The dual-cancellation Proposition makes the repair algebraic; it is presented as an adaptation of standard Sinkhorn structure rather than a new normalization, and it does not imply a UOT-over-BOT advantage (balanced vs. unbalanced conditional decoding differ by  $-0.0003$  with a CI including zero).

**RC-UOT-Q** adds event-backed quotient groups and coverage qualification with abstention. On the Celer benchmark (7,296 anchor pairs; 7,128 frozen flow labels), its coverage machinery reproduces its own event-backed quotient grouping exactly on the 122 covered holdout pairs (a definitional

consistency check; Supplement Table S1) and improves precision/F1 over adapted diagnostic baselines on a development-sealed holdout (calibration-asymmetry caveat, Section IV-E); original Connector native behavior is reported separately (Section IV-C1). We release the **Cross AML** prototype (`tools/cross_aml/`) for evidence-to-report investigator triage.

Taken together, CSFFC, RC-UOT, RC-UOT-Q, and the Cross AML prototype provide an evidence-aware, auditable, coverage-qualified path from transaction hashes to scoped cross-chain fund-flow correspondence hypotheses. Method performance on a fully adequate independent real-world corpus remains unestablished: the data-only audit (Section IV-G) established real flow-level fan-out outside the development corpus, but the corpus did not meet the preregistered adequacy gate and no method was executed on it, so no external-performance claim is made.

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